

Fig. 2: Actual-size PCB layout of the USB interface signal monitor

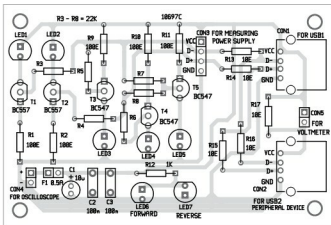


Fig. 3: Components layout for the PCB

PARTS LIST

Semiconductors:

- T1, T2 - BC557 npn transistor
- T3-T5 - BC547 npn transistor
- LED1-LED7 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R9-R11 - 100-ohm
- R3-R8 - 22-kilo-ohm
- R12 - 1-kilo-ohm
- R13-R17 - 10-ohm

Capacitors:

- C1 - 10 μ F, 16V electrolytic
- C2, C3 - 100nF ceramic disk

Miscellaneous:

- CON1, CON2 - USB connector A type
- CON3 - 4-pin connector
- CON4, CON5 - 2-pin connector
- F1 - 0.5A fuse with holder
- USB cable

current flowing through the LEDs. You can increase the resistor values from 100 ohms to 500 ohms as per your requirements.

Resistors R3 through R8 are used to limit the base current of transistors T1 through T5 and the load of DATA+ and DATA-.

These resistors should have the maximal possible values in order

to reduce the USB load. The exact values of these resistors depends on the DC gain of the transistors and the current required for the LEDs.

Power supply. The circuit operates in the full power supply range of the USB interface. If polarity of the supply from the USB connector is reversed, LED7 glows. Otherwise, it remains off and LED6 glows during normal (forward) operation.

Current measurement. Resistor R17 and connector CON5 are used to measure the current of the interface. For example, if the power supply current is 0.3A, the voltage drop across resistor R17 will be 30mV. Use a digital voltmeter or oscilloscope to measure the voltage drop across R17.

Oscilloscope. You can connect the oscilloscope to 4-pin header CON3 or to 2-pin header CON5. Also, you can use connector CON4 to connect an oscilloscope, voltmeter or module operating from +5V supply.

Adjustment. The circuit does not need any special adjustment to operate. However, in some cases, you may need to change the values of some resistors, and use transistors with high DC gain and LEDs with low forward voltage drop, but that's very rare.

Construction and testing

An actual-size PCB layout of the USB interface monitor is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable box such that all the connectors are easily accessible. Mount the LEDs (LED1 through LED7) on the front side of the box for visual indication of signals. **EFY**



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